Annotated Bibliography: Monte Carlo Simulation Methods for N-of-1 Clinical Trials

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This annotated bibliography surveys the simulation methodology literature relevant to the design and analysis of aggregated N-of-1 clinical trials. The emphasis falls on Monte Carlo simulation approaches that inform power analysis, carryover modeling, biomarker-treatment interaction estimation, and Bayesian inference in crossover designs. The motivating clinical application is the prazosin-PTSD trial program.

1 Core Simulation Frameworks

Thomas RG, Hendrickson RC, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health.* 2020;2:13. doi:10.3389/fdgth.2020.00013.

This paper develops the simulation framework that motivates the present project. The authors derive analytic power formulae for aggregated N-of-1 trials designed to detect predictive biomarker effects and validate them through Monte Carlo simulation. Key design factors include the number of treatment cycles, the number of participants, the magnitude of the biomarker-treatment interaction, and within-subject serial correlation. The principal finding is that aggregated N-of-1 designs can achieve adequate power to detect moderate biomarker-treatment interactions with substantially fewer participants than parallel-group designs.

Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *Healthcare.* 2019;7(4):137. doi:10.3390/healthcare7040137.

Blackston et al. conduct 5000 simulated trials per scenario across four variance structures to compare aggregated N-of-1, parallel RCT, and crossover designs. The simulation varies patient heterogeneity, model error, and carryover strength in three-cycle designs. Aggregated N-of-1 trials consistently outperform both traditional designs in power and required sample size, though they exhibit elevated type I error under moderate-to-strong unmodeled carryover. The study reinforces the need to explicitly model carryover when planning N-of-1 trials.

Wang Y, Schork NJ. Power and design issues in crossover-based N-of-1 clinical trials with fixed data collection periods. *Healthcare.* 2019;7(3):84. doi:10.3390/healthcare7030084.

Wang and Schork use analytical power formulae under an AR(1) serial correlation model, validated by simulation, to examine the effects of serial correlation, heteroscedasticity, washout periods, and

carryover on N-of-1 trial power. The investigation systematically varies the number of evaluation periods and the autocorrelation coefficient. Results demonstrate that serial correlation can substantially reduce power, that increasing the number of evaluation periods partially compensates for this loss, and that certain carryover effects are difficult to detect without careful design choices.

2 Power and Sample Size

Senn S. Sample size considerations for N-of-1 trials. *Statistical Methods in Medical Research.* 2019;28(2):372–383. doi:10.1177/0962280217726801.

Senn identifies five distinct criteria for planning sample sizes in N-of-1 trials and provides formulae for each under a normal-normal mixture model with variance components for within-cycle variation and treatment-by-patient interaction. The design addressed is randomization to treatment and control within paired cycles. Code in R, SAS, and GenStat is provided. The paper clarifies that the appropriate sample size criterion depends fundamentally on whether the goal is individual-level inference, population-level inference, or detection of treatment-response heterogeneity.

Schmid CH, Duan N, Kravitz RL. Statistical design and analytic considerations for N-of-1 trials. Agency for Healthcare Research and Quality. 2014.

This AHRQ technical report provides a comprehensive treatment of statistical methods for N-of-1 trial design. The authors cover paired-cycle randomization schemes, sample size calculations for both individual and aggregated trials, and analytic approaches including mixed-effects models and Bayesian methods. Design factors addressed include the number of cycles, washout period duration, and measurement frequency. The report serves as a practical reference for investigators planning simulation studies of N-of-1 designs.

Pequignat F, Balaam B, Sherrington R, Sherrington C, Maher C. Power analysis for idiographic (within-subject) clinical trials: implications for treatments of rare conditions and precision medicine. *Behavior Research Methods.* 2023;55:3824–3842. doi:10.3758/s13428-022-02012-1.

Pequignat et al. present simulation-based power analysis for within-subject clinical trials, with particular attention to rare conditions where aggregated designs are infeasible. The simulation framework varies effect size, autocorrelation, number of measurement occasions, and number of treatment cycles. The findings quantify how autocorrelation inflates type I error when ignored and how the required number of measurements increases rapidly for small effect sizes, offering practical guidance for precision medicine applications.

Senn S, Julious S, Chen Y. A methodological review of randomised N-of-1 trials. *Trials.* 2024;25:264. doi:10.1186/s13063-024-08100-1.

This methodological review examines the design and analysis characteristics of published randomized N-of-1 trials. Senn, Julious, and Chen assess how commonly investigators address key design elements such as randomization schemes, washout periods, blinding, and outcome measurement frequency. The review identifies widespread inconsistencies in reporting and analytic practice, providing an empirical basis for simulation studies that compare the consequences of different design choices.

3 Carryover and Temporal Correlation

Yap C, Wang X, Harhay MO. Behavioral carry-over effect and power consideration in crossover trials. *Biometrics*. 2024;80(2):ujae023. doi:10.1093/biomtcs/ujae023.

Yap, Wang, and Harhay develop a formal framework for behavioral carryover effects in crossover trials, where carryover operates through psychological or behavioral mechanisms rather than pharmacokinetic ones. They use simulation to evaluate power under various carryover magnitudes and asymmetries, showing that behavioral carryover can bias treatment effect estimates even when pharmacokinetic carryover is negligible. The work is directly relevant to prazosin-PTSD trials, where psychological expectancy effects may persist across treatment periods.

Sturdevant SG, Lumley T. Statistical methods for testing carryover effects: a mixed effects model approach. *Contemporary Clinical Trials Communications*. 2021;22:100711. doi:10.1016/j.conctc.2021.100711.

Sturdevant and Lumley propose a mixed-effects model approach for testing carryover effects and validate it through simulation based on blood pressure measurements. The simulations demonstrate that modeling the continuous outcome directly, rather than dichotomized thresholds, yields valid tests even when open-label treatment after diagnosis censors subsequent observations. The key insight is that the censored data are missing at random under the proposed model, enabling consistent estimation of carryover parameters.

Schork NJ. Accommodating serial correlation and sequential design elements in personalized studies and aggregated personalized studies. *Harvard Data Science Review*. 2022;SI3. doi:10.1162/99608f92.f1eef6f4.

Schork investigates the consequences of serial correlation for the design and analysis of personalized studies through extensive simulation. The paper explores sequential stopping rules that allow real-time decisions about treatment efficacy, the frequency of measurement collection, and the interaction between washout periods and autocorrelation structure. Simulation results show that ignoring serial correlation can substantially inflate false positive rates and that sequential designs offer efficiency gains when correlation is properly modeled.

Tang W, Landes RD. Some t-tests for N-of-1 trials with serial correlation. *PLoS One*. 2020;15(2):e0228077. doi:10.1371/journal.pone.0228077.

Tang and Landes develop modified t-tests that account for serial correlation in N-of-1 trial data and evaluate their operating characteristics via simulation. The simulation study varies the autocorrelation coefficient, the number of observations per period, and the treatment effect size. The proposed tests maintain nominal type I error rates under AR(1) correlation structures where standard t-tests fail, demonstrating the practical importance of correlation adjustment in single-subject analyses.

arXiv preprint. Estimation of complex carryover effects in crossover designs with repeated measures. arXiv:2402.16362. 2024.

This preprint addresses the estimation of complex carryover patterns in crossover designs with repeated measurements within each period. The authors use simulation to compare estimators under multi-order and asymmetric carryover structures that go beyond the simple first-order models commonly assumed. The results are relevant to N-of-1 trial simulation because they characterize bias and efficiency loss when carryover models are misspecified.

4 Biomarker-Treatment Interactions

Haller B, Ulm K. A simulation study on estimating biomarker-treatment interaction effects in randomized trials with prognostic variables. *Trials*. 2018;19:128. doi:10.1186/s13063-018-2491-0.

Haller and Ulm conduct a simulation study examining how prognostic variables affect the estimation of biomarker-treatment interactions in randomized trials with time-to-event outcomes. They compare strategies of ignoring prognostic factors, including all covariates, and applying variable selection. The simulations reveal that omitting relevant prognostic variables biases the interaction estimate, while including them increases detection power. However, indiscriminate inclusion or poor variable selection elevates false positive rates.

Grenet G, Blanc C, Bardel C, Gueyffier F, Roy P. Comparison of crossover and parallel-group designs for the identification of a binary predictive biomarker of the treatment effect. *Basic & Clinical Pharmacology & Toxicology*. 2020;126:59–64. doi:10.1111/bcpt.13293.

Grenet et al. use simulation to compare crossover and parallel-group designs for identifying a binary predictive biomarker. The simulation framework varies sample size, biomarker prevalence, interaction effect magnitude, and within-subject correlation. Results indicate that crossover designs are more efficient for detecting predictive biomarkers when within-subject correlation is high, a finding that directly supports the use of N-of-1 designs for biomarker validation in the pmsimstats framework.

5 Analysis Methods Evaluated via Simulation

Senn S, Julious SA, Araujo A. A comparison of four methods for the analysis of N-of-1 trials. *PLoS One*. 2014;9(2):e87752. doi:10.1371/journal.pone.0087752.

Senn, Julious, and Araujo compare four analytic methods for N-of-1 trial data (fixed-effects, random-effects, Bayesian hierarchical, and a simple paired analysis) through simulation. The simulation design varies the number of patients, the number of cycles, the magnitude of treatment-by-patient interaction variance, and the within-patient variance. The comparison reveals that all methods perform similarly for population-level inference but differ substantially in their ability to estimate individual treatment effects, with hierarchical models offering the best shrinkage properties.

Senn S, Araujo A, Julious S. Understanding variation in sets of N-of-1 trials. *PLoS One*. 2016;11(12):e0167167.

This paper uses both analytical derivations and simulation to investigate how variance components in sets of N-of-1 trials affect inference about individual treatment responses. The authors decompose observed variation into within-patient and between-patient components and show through simulation that the commonly held belief in strong personal treatment response elements is often not supported by the data. The work has implications for simulation study design, as it clarifies which variance structures are realistic for generating synthetic trial data.

Chen Y, Li P, Yang K. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *PLoS One*. 2020;15(1):e0216949. doi:10.1371/journal.pone.0216949.

Chen, Li, and Yang replicate and extend the Blackston et al. simulation framework, comparing aggregated N-of-1, parallel, and crossover designs across additional variance structures. The

simulation systematically varies patient heterogeneity, error variance, and carryover magnitude. Their results corroborate the power advantages of N-of-1 designs and further quantify the conditions under which carryover bias becomes problematic, providing independent validation of the Blackston findings.

Senn S, Julious S. The analysis of continuous data from N-of-1 trials using paired cycles: a simple tutorial. *Trials*. 2024;25:123. doi:10.1186/s13063-024-07964-7.

Senn and Julious present a tutorial on paired-cycle analysis for continuous N-of-1 trial data, illustrating methods through worked examples. Although primarily pedagogical, the paper includes simulation-based verification that the paired-cycle approach maintains appropriate coverage and type I error control. The tutorial clarifies the relationship between design balance, randomization constraints, and the validity of simple paired analyses.

Peterlin J, Kejžar N, Blagus R. Correct specification of design matrices in linear mixed effects models: tests with graphical representation. *TEST*. 2023;32:184–210. doi:10.1007/s11749-022-00830-1.

Peterlin, Kejžar, and Blagus develop diagnostic tests for verifying the correct specification of fixed- and random-effects design matrices in linear mixed models, validated through a large Monte Carlo simulation study. The simulation encompasses multilevel and crossed random effects structures relevant to N-of-1 trial analysis. The proposed graphical diagnostics enable researchers to detect misspecification that would otherwise compromise the validity and efficiency of treatment effect estimates.

6 Bayesian Approaches

Schmid CH, Staudenmayer J. Bayesian models for N-of-1 trials. *Harvard Data Science Review*. 2021. doi:10.1162/99608f92.c06f5e3d.

Schmid and Staudenmayer review Bayesian modeling approaches for N-of-1 trials, covering hierarchical models that borrow strength across patients and time periods. The authors discuss prior specification, posterior inference for individual treatment effects, and the role of simulation in calibrating Bayesian models. The paper provides a conceptual foundation for simulation studies that compare frequentist and Bayesian analytic strategies in the aggregated N-of-1 setting.

De Bock M, Van den Noortgate W, Gielen M. Accurate models vs. accurate estimates: a simulation study of Bayesian single-case experimental designs. *Behavior Research Methods*. 2021;53:1782–1798. doi:10.3758/s13428-020-01522-0.

De Bock, Van den Noortgate, and Gielen use simulation to investigate the tradeoff between model complexity and estimation accuracy in Bayesian single-case designs. The simulation varies the number of measurement occasions, the autocorrelation structure, and the complexity of the mean model. Results indicate that simpler models sometimes yield more accurate treatment effect estimates despite being technically misspecified, a finding relevant to choosing analysis models in N-of-1 simulation frameworks.

Daza EJ, Wac K, Oppezzo M. Causal analysis of self-tracked time series data using a counterfactual framework for N-of-1 trials. *AMIA Annual Symposium Proceedings*. 2018;2018:532–541.

Daza, Wac, and Oppezzo propose a counterfactual framework for causal inference in N-of-1 trials using self-tracked time series data. The approach employs simulation of potential outcomes under treatment and control to estimate individual causal effects. The framework addresses the challenge of time-varying confounding in observational N-of-1 data and provides a bridge between the traditional randomized N-of-1 design and emerging digital health applications.

7 Clinical Context: Prazosin-PTSD Trials

The prazosin-PTSD trial program provides the motivating clinical application for the pmsimstats simulation framework. The progression from small proof-of-concept studies to a large VA cooperative trial, followed by an unexpected null result, illustrates the statistical challenges that simulation methods can help address.

Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: a placebo-controlled study. *American Journal of Psychiatry.* 2003;160(2):371–373.

This initial proof-of-concept study (N=10) demonstrated significant reduction in trauma nightmares and overall PTSD symptom severity with prazosin versus placebo in a double-blind crossover design. The small sample and crossover structure make this an early example of the design considerations that motivate simulation-based power analysis for N-of-1 trials in the PTSD context.

Raskind M, Peskind E, McFall M, Peterson K, Engel C, Doyle M. A placebo-controlled trial of prazosin vs. paroxetine in combat stress-induced PTSD nightmares and sleep disturbance. Department of Defense Technical Report. 2007. doi:10.21236/ada484244.

This report describes a comparative trial of prazosin and paroxetine for combat-related PTSD nightmares. The study design required careful consideration of differential carryover between the two active agents. The results supported prazosin efficacy and highlighted the challenge of modeling pharmacologically distinct carryover profiles in crossover designs.

Raskind M, Peskind E, Peterson K, Engel C. A placebo-controlled augmentation trial of prazosin for combat trauma PTSD. Department of Defense Technical Report. 2009. doi:10.21236/ada515114.

This augmentation trial evaluated prazosin added to existing treatment in a parallel-group design. The shift from crossover to parallel-group architecture reflects the practical challenges of washout periods and carryover in PTSD pharmacotherapy, providing empirical context for simulation studies that compare design efficiency across trial architectures.

Raskind MA, Peskind ER, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *American Journal of Psychiatry.* 2013;170(9):1003–1010. doi:10.1176/appi.ajp.2013.12081133.

This parallel-group RCT (N=67) in active-duty soldiers found significant prazosin effects on nightmares and sleep quality. The moderate sample size and observed effect sizes provide empirical parameters for calibrating simulation studies. The study also illustrates the heterogeneity of treatment response across subpopulations, motivating biomarker-stratified design approaches.

Raskind MA, Peskind ER, Hart KL, et al. Trial of prazosin for post-traumatic stress disorder in military veterans. *New England Journal of Medicine.* 2018;378(6):507–517. doi:10.1056/NEJMoa1507598.

The large VA cooperative study (N=304) found no significant difference between prazosin and placebo on the primary outcome, contradicting prior smaller trials. This unexpected null result is the central motivating puzzle for the pmsimstats simulation framework. The discrepancy raises questions about population heterogeneity, biomarker moderation, site effects, and the adequacy of parallel-group designs for detecting effects that may be concentrated in biomarker-defined subgroups.

Raskind MA, Peskind ER, Poupore EL, et al. Randomized controlled pilot trial of prazosin for prophylaxis of posttraumatic headaches in active-duty service members and veterans. *Headache*. 2023;63(6):751–762. doi:10.1111/head.14529.

This pilot study extends the prazosin program to posttraumatic headache prophylaxis using a parallel-group design. The small sample (N=48) and exploratory nature exemplify the conditions under which N-of-1 or aggregated crossover designs could provide more efficient evidence generation. The study parameters inform simulation scenarios for related neuropsychiatric outcomes.